

12.1

$A = 150 \text{ sm} = 1,5 \text{ m}$	$x = A \sin(\omega t + \varphi_0)$	$\omega = 2\pi/4 = \pi/2 \text{ rad/s}$
$\varphi = 45^\circ$	$\omega = 2\pi/T$	$x = 1,5 \sin(\pi/2 \cdot t + 45^\circ)$
$T = 4 \text{ s}$		Grafik chiziladi

12.2

$A = 0,1 \text{ m}$	$x = A \sin(\omega t + \varphi_0)$	$\omega = 2\pi/4 = \pi/2 \text{ rad/s}$
$T = 4 \text{ s}$	$\omega = 2\pi/T$	$x = 0,1 \sin(\pi/2 \cdot t + \pi/2)$
$\varphi_0 = \pi/2$		Grafik chiziladi

12.3

$A = 50 \text{ mm} = 0,05 \text{ m}$	$x = A \sin(\omega t + \varphi_0)$	1) $\omega = 2\pi/4 = \pi/2 \text{ rad/s}$
$T = 4 \text{ s}$	$\omega = 2\pi/T$	$x = 0,05 \sin(\pi/2 \cdot t)$
$t_1 = 0$		2) $t = 0: x_0 = 0$
$t_2 = 1,5 \text{ s}$		$t = 1,5 \text{ s}: x = 0,05 \sin(\pi/2 \cdot 1,5) = 0,05 \sin(0,75\pi) \approx 0,035 \text{ m}$
		3) Grafik chiziladi

12.11

$x = \sin(\pi t/6)$	$v = dx/dt = (\pi/6)\cos(\pi t/6)$	$v_{\max} = \pi/6 \text{ sm/s}$
	$a = d^2x/dt^2$	$a_{\max} = (\pi/6)^2 = \pi^2/36 \text{ sm/s}^2$
		Grafik vaqtga bog'liq ch...

12.12

$T = 2 \text{ s}$	$x = A \sin(\omega t + \varphi_0)$	$\omega = 2\pi/2 = \pi \text{ rad/s}$
$A = 50 \text{ mm} = 5 \text{ sm}$	$v = A\omega \cos(\omega t + \varphi_0)$	$\sin(\pi t + \pi/2) = 2,5/5 = 0,5$
$\varphi_0 = \pi/2$		$\pi t + \pi/2 = \pi/6 \rightarrow t = -1/3 \text{ s (not)}$
$x = 25 \text{ mm} = 2,5 \text{ sm}$		$yoki \pi t + \pi/2 = 5\pi/6 \rightarrow t = 1/3 \text{ s}$
		$v = 5\pi \cos(\pi/3 + \pi/2) \approx -13,6 \text{ s}$

12.4

$\varphi_1 = 0$	$x = A \sin(\omega t + \varphi_0)$	$\omega = 2\pi/8 = \pi/4 \text{ rad/s}$
$\varphi_2 = \pi/3$	$\omega = 2\pi/T$	$x_1 = 5 \sin(\pi/4 \cdot t + 0)$
$\varphi_3 = \pi/4$		$x_2 = 5 \sin(\pi/4 \cdot t + \pi/3)$
$\varphi_4 = \pi/2$		$x_3 = 5 \sin(\pi/4 \cdot t + \pi/4)$
$\varphi_5 = 2\pi$		$x_4 = 5 \sin(\pi/4 \cdot t + \pi/2)$
$A = 5 \text{ sm}$		$x_5 = 5 \sin(\pi/4 \cdot t + 2\pi)$
$T = 8 \text{ s}$		Grafiklar chiziladi

12.5

$A_1 = A_2 = 2 \text{ sm}$ $T_1 = T_2 = 8 \text{ s}$ $\varphi_1 = \pi/4$ $\varphi_2 = \pi/3$ $\varphi_3 = \pi$ $\varphi_4 = 2\pi$	$x = A \sin(\omega t + \varphi_0)$	Har bir boshlang'ich faza uchun grafik chiziladi: $x_1 = 2 \sin(\pi/4 \cdot t + \pi/4)$ $x_2 = 2 \sin(\pi/4 \cdot t + \pi/3)$ $x_3 = 2 \sin(\pi/4 \cdot t + \pi)$ $x_4 = 2 \sin(\pi/4 \cdot t + 2\pi)$
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12.6

t = 24 x = A sin(ωt Boshlang'ich fazadan qancha vaqt o'tgach muvozanat vaziyatdan yarim
s + φ₀) amplitudaga teng siljiydi? Bu yerda fazaviy shart qo'llanadi
A ω = 2π/T



12.13

$v_{\max} = 49,3 \text{ sm/s}^2$	$v_{\max} = A\omega$	$\omega = 2\pi/2 = \pi \text{ rad/s}$
$T = 2 \text{ s}$	$\omega = 2\pi/T$	$A = v_{\max}/\omega = 49,3/\pi \approx 15,7 \text{ sm}$
$x = 25 \text{ mm} = 2,5 \text{ sm}$	$a = -\omega^2 x$	$a = -\pi^2 \cdot 2,5 \approx -24,7 \text{ sm/s}^2$

12.14

$\varphi_0 = \pi/2$	$v^2 = \omega^2(A^2 - x^2)$	$\omega^2 A^2 = v_1^2 + \omega^2 x_1^2$
$v_1 = 2,4 \text{ sm/s}$	$E = \frac{1}{2} m \omega^2 A^2$	$2,4^2 = \omega^2 A^2$
$v_2 = 2 \text{ sm/s (3 sm da)}$		$2^2 = \omega^2 (A^2 - 3^2)$
$v_3 = 2,8 \text{ sm/s (esa 2 sm/s da)}$		Tenglamalarni yechib A va ω topiladi

12.15

$m = 1,6 \cdot 10^{-2} \text{ kg}$	$F = -kx$	$\omega = \pi/2 \text{ rad/s}, A = 0,1 \text{ m}$
$x = 0,1 \sin(\pi t/2 + \pi/4)$	$k = m\omega^2$	$k = 1,6 \cdot 10^{-2} \cdot (\pi/2)^2 \approx 0,0395 \text{ N/m}$
	$E = \frac{1}{2}kA^2$	$F_{\max} = kA = 0,00395 \text{ N}$
		Grafik chiziladi

12.16

$m = 10 \text{ g} = 0,01 \text{ kg}$	$E = \frac{1}{2} m \omega^2 A^2$	$A = 5 \text{ sm} = 0,05 \text{ m}, \omega = \pi/5$
$x = 5 \sin(\pi t/5 + \pi/4)$	$v_{\text{max}} = A \omega$	$v_{\text{max}} = 0,05 \cdot \pi/5 = \pi/100 \text{ m/s}$
		$E = \frac{1}{2} \cdot 0,01 \cdot (\pi/5)^2 \cdot 0,05^2 \approx 4,9 \cdot 10^{-6} \text{ J}$

12.7

$\varphi_0 = \pi/2$	$x = A \sin(\omega t + \varphi_0)$ $v = A\omega \cos(\omega t + \varphi_0)$ $a = -A\omega^2 \sin(\omega t + \varphi_0)$	Boshlang'ich fazasi $\pi/2$ bo'lganda: $x_0 = A, v_0 = 0, a_0 = -A\omega^2$ Maksimal tezlik $\omega = 2\pi\nu$ dan topiladi
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12.8

$$\begin{array}{lll} x = 7 \sin 0,5t & x = A \sin(\omega t) & A = 7 \text{ sm} \\ v = A \omega \cos(\omega t) & \omega = 0,5 \text{ rad/s} & T = 2\pi/\omega = 2\pi/0,5 = 4\pi \text{ s} \\ & v_{\max} = A\omega = 7 \cdot 0,5 = 3,5 \text{ sm/s} \end{array}$$

12.9

$$\begin{aligned} A &= 5 \text{ sm} & v &= A\omega \cos(\omega t + \varphi_0) & \omega &= 2\pi/4 = \pi/2 \text{ rad/s} \\ T &= 4 \text{ s} & v_{\max} &= A\omega & v_{\max} &= 5 \cdot \pi/2 = 2,5\pi \approx 7,85 \text{ sm/s} \\ & & \omega &= 2\pi/T \end{aligned}$$

12.10

$x = 2\sin(\pi t/2 + \pi/4)$ $x = A \sin(\omega t + \varphi_0)$ 1) $A = 2 \text{ sm}, \omega = \pi/2, T = 4 \text{ s}$
 $v = dx/dt$ 2) $v_{\max} = A\omega = 2 \cdot \pi/2 = \pi \text{ sm/s}$
 $a = d^2x/dt^2$ 3) $a_{\max} = A\omega^2 = 2 \cdot (\pi/2)^2 = \pi^2/2 \text{ sm/s}^2$



12.17			12.22		12.26
B:	F:	Y:	B:	F:	Y:
m = 16 g = 0,016 kg x = 2sin(πt/4 + π/6)	E _k = ½mv² E _p = ½kx² E = E _k + E _p	A = 2 sm, ω = π/4 k = mω² = 0,016 · (π/4)² Grafiklar vaqtga bog'liq chiziladi	l = 2 m α = 4°	T = 2π√(l/g) g = 9,8 m/s²	T = 2π√(2/9,8) = 2,84 s Kichik burchak uchun garmonik tebranish
12.19			12.23		
B:	F:	Y:	B:	F:	Y:
x ₁ = A/4 x ₂ = A/2 x ₃ = A	E _k = ½mω²(A² - x²) E _p = ½mω²x² E = ½mω²A²	1) E _{k1} /E _{p1} = (A² - A²/16)/(A²/16) = 15 2) E _{k2} /E _{p2} = (A² - A²/4)/(A²/4) = 3 3) x = A da E _k = 0, E _p = E	m = 10 kg k = 1 kg l = 1,5 m	T = 2π√(l/g)	T = 2π√(1,5/9,8) ≈ 2,46 s Vertikal tebranish davrini aniqlang
12.20			12.24		
B:	F:	Y:	B:	F:	Y:
W = 3·10 ⁻³ J k = 1,5·10³ N/m T = 2 s φ = 60°	E = ½kA² x = A sin(ωt + φ ₀)	A = √(2E/k) = √(2 · 3·10 ⁻³ /1,5·10³) = 0,063 m ω = 2π/2 = π x = 0,063 sin(πt + 60°)	E _k = E _p A = 5 sm	E _k = ½mω²(A² - x²) E _p = ½mω²x²	E _k = E _p → A² - x² = x² x = A/√2 = 5/√2 ≈ 3,54 sm k topiladi
12.21			12.25		
B:	F:	Y:	B:	F:	Y:
A = 2 sm = 0,02 m W = 3·10 ⁻⁴ J	E = ½mω²A² ω² = 2E/(mA²)	W = 2,25·10 ⁻⁴ N·m da k = 2E/A² = 2 · 3·10 ⁻⁴ /0,02² ≈ 1,5 N/m To'la energiya topiladi	Vertikal prujina yuk osilgan	T = 2π√(m/k) k parallel va ketma-ket	Parallel: k _p = k ₁ + k ₂ Ketma-ket: 1/k _k = 1/k ₁ + 1/k ₂ Davrlar taqqoslanadi
12.30					
B:	F:	Y:			
Tomonga 8 s Amplituda 0,02 m φ = π/6	x = A sin(ωt + φ ₀) ω = 2π/T	ω = 2π/8 = π/4 x = 0,02 sin(π/4 · t + π/6) Grafik chiziladi			
12.31			12.32		
B:	F:	Y:	B:	F:	Y:
x ₁ = 0,02sin(5πt + π/6) x ₂ = 0,03sin(5πt + π/3)	x = x ₁ + x ₂ A = √(A ₁ ² + A ₂ ² + 2A ₁ A ₂ cosΔφ)	Δφ = π/3 - π/6 = π/6 A = √(0,02² + 0,03² + 2·0,02·0,03·cos(π/6)) ≈ 0,049 m	Ikki tomonga T ₁ = T ₂ A ₁ = A ₂	x = x ₁ + x ₂ Fazalar farqi	Natija amplituda va fazaga qarab aniqlanadi
			12.33		

B:	F:	Y:
Vertikal prujina mis sharcha r radius	T = 2π√(m/k) m = ρV V = 4πr³/3	Tebranish davri: T = 2π√(4πr³ρ/3k) Yuk tufayli

B:	F:	Y:
Vertikal prujina T ₁ = 0,5 s T ₂ qo'shilganda	T = 2π√(m/k)	T ₁ ² = 4π²m ₁ /k T ₂ ² = 4π²m ₂ /k Qo'shilgan yuk tufayli davr o'zgaradi

B:	F:	Y:
l = 1 mm = 0,001 m m = 0,5 kG E = 0,3 kG/mm²	T = 2π√(m/k) k = EA/l	k = 0,3 · 10⁹ · A / 0,001 T = 2π√(0,5/k) Vertikal tebranish davri

B:	F:	Y:
P = 0,2 kG T = 3,4 s d = 1 sm	T = 2π√(m/k) k = P/Δl	Δl topiladi: k = P/Δl T² = 4π²m/k P · ↓ etrlardan Δl hisoblanadi

B:	F:	Y:
x ₁ = 4sin(πt) x ₂ = 3sin(πt + π/2)	A = √(A ₁ ² + A ₂ ² + 2A ₁ A ₂ cosΔφ)	Δφ = π/2 A = √(16 + 9 + 0) = 5 sm 1) Tenglamalar berilgan 2) Natija topilgan 3) Vektor diagrammasi

12.1 Agar 1 min.da 150 tebranish bo'lib, tebranishlarning boshlang'ich fazasi 45° ga teng va amplitudasi 5 sm bo'lgan garmonik tebranma harakat tenglamasini yozing. Shu harakat grafigini chizing.

12.2. Amplitudasi 0,1 m, davri 4 sek. va boshlang'ich fazasi nolga teng bo'lgan garmonik tebranma harakat tenglamasini yozing.

12.3. Garmonik tebranishlar amplitudasi 50 mm, davri 4 sek. va boshlang'ich fazasi $\frac{\pi}{4}$. 1) Mazkur tebranishning tenglamasini yozing. 2) $t=0$

va $t=1,5$ sek bo'ganda tebranayotgan nuqtaning muvozanat vaziyatdan siljishini toping. 3) Bu harakat grafigini chizing.

12.4. Tebranishlarning boshlang'ish fazasi: 1) 0, 2) $\frac{\pi}{2}$ 3) π 4) $\frac{3}{4}\pi$ 5) 2π

bo'lgan garmonik tebranma harakat tenglamasini yozing. Tebranishlar amplitudasi 5 sm va tebranishlar davri 8 sek. Bu hollarning hammasi uchun tebranishlar grafigini chizing.

12.5. Birday amplitudada ($A_1=A_2=2$ sm) va birday davrga ($T_1=T_2=8$ sek) ega bo'lgan, lekin fazalarning farqi: 1) $\frac{\pi}{4}$, 2) $\frac{\pi}{2}$ 3) π , 4) 2π bo'lgan ikki garmonik tebranishni bitta grafikda chizing.

12.6. Garmonik tebranayotgan nuqta harakat boshlanishidan qancha vaqt o'tgach muvozanat vaziyatdan yarim amplitudaga teng siljiydi? Tebranish davri 24 sek, boshlang'ich faza nolga teng.

12.7. Garmonik tebranishlarning bosglang'ich fazasi nolga teng. Davrning qanday qismini o'tgach nuqtaning tezligi maksimal tezligining yarmiga teng bo'ladi?

12.8. $x=7\sin 0,5t$ tenglama bo'yicha tebranma harakat qilinayotgan nuqta harakat boshlanganidan qancha vaqt o'tgach muvozanat vaziyatdan maksimal siljishga qadar yo'lni o'tadi.

12.9. Garmonik tebranish amplitudasi 5 sm, davri 4 sekunda teng. Tebranayotgan nuqtaning maksimal tezligini toping.

12.10. Nuqtaning harakat tenglamasi $x=2\sin(\frac{\pi}{2}t+\frac{\pi}{4})$ sm ko'rinishda berilgan. 1) Tebranishlar davrini, 2) nuqtaning maksimal tezligini, 3) uning maksimal tezlanishini toping.

12.11. Nuqtaning harakat tenglamasi $x=\sin\frac{\pi}{6}t$ ko'rinishda berilgan. Maksimal tezlik va maksimal tezlanishlarga erishiladigan vaqt paytlarini toping.

12.12. Nuqta garmonik tebranadi. Tebranishlar davri 2 sek, amplitudasi 50 mm, boshlang'ich fazasi nolga teng. Nuqtaning muvozanat vaziyatidan 25mm ga siljigan paytdagi tezligini toping.

12.13. Nuqtaning maksimal tezlanishi $49,3$ sm/sec², tebranishlar davri 2 sek va nuqtaning muvozanat vaziyatidan 25 mm ga siljigan boshlang'ich vaqt paytidagi garmonik tebranishlarning harakat tenglamasini yozing.

12.14. Garmonik tebranishlarning bosglang'ich fazasi nolga teng. Nuqta muvozanat vaziyatidan 2,4 sm siljiganda nuqtaning tezligi 3 sm/sec bo'ladi, 2,8 sm ga siljiganida esa tezligi 2 sm/sec bo'ladi. Shu tebranishlar amplitudasi va davrini toping.

12.15. $m=1,6\cdot 10^{-2}$ kg massali moddiy nuqtaning tebranish tenglamasi $x=0,1\sin(\frac{\pi}{8}t-\frac{\pi}{4})$ ko'rinishga ega. Nuqtaga ta'sir etuvchi F kuchning (bitta davr chegarasida) t vaqtga bog'liqlik grafigini chizing. Maksimal kuch qiymatini toping.

12.16. Massasi 10 g bo'lgan moddiy nuqta $x=5\sin(\frac{\pi}{5}t+\frac{\pi}{4})$ sm tenglama bo'yicha tebranadi. Nuqtaga ta'sir etuvchi maksimal kuchni va tebranayotgan nuqtaning to'la energiyasini toping.

12.17. Massasi 16 g bo'lgan moddiy nuqtaning tebranish tenglamasi $x=2\sin(\frac{\pi}{4}t+\frac{\pi}{4})$ sm ko'rinishga ega. Nuqtaning kinetik, potensial va to'la energiyasining bitta davr chegarasida vaqtga bog'liqlik grafigini chizing.

12.19. Nuqtaning muvozanat vaziyatidan siljishi: 1) $x = \frac{A}{4}$, $x = \frac{A}{2}$, 3) $x = A$ ni tashkil qiladigan (A -tebranish amplitudasi) paytlar uchun garmonik tebranayotgan nuqta kinetik energiyasining uning potensial energiyasiga bo'lgan nisbati nimaga teng?

12.20. Garmonik tebranma harakatda bo'lgan jismning to'la energiyasi $3 \cdot 10^{-5}$ J, jismga ta'sir etuvchi maksimal kuch $1,5 \cdot 10^{-3}$ N ga teng. Tebranish davri 2 sek va boshlang'ich fazasi 60° bo'lsa, shu jismning harakat tenglamasini yozing.

12.21. Moddiy nuqtaning garmonik tebranishlar amplitudasi $A=2$ sm, tebranishlarning to'la energiyasi $W=3 \cdot 10^{-7}$ J. Muvozanat vaziyatidan qancha siljiganda tebranayotgan nuqtaga $F=2,25 \cdot 10^{-5}$ N kuch ta'sir etadi?

12.22. Uzunligi 2 m bo'lgan ipga osilgan sharchani 4° burchakka og'dirilib, tebranishi kuzatiladi. Tebranishni so'nmaydigan garmonik tebranish deb faraz qilib, sharcha muvozanat vaziyatidan o'tayotgandagi tezligini toping. Mexanika tenglamalaridan sharchaning muvozanatlik vaziyatidan o'tish tezligini topib, olingan yechimni tekshirib ko'ring.

12.23. Prujinaga 10 kg yuk osilgan. Prujina 1 kg kuch ta'sirida 1,5 sm cho'zilishi mumkin bo'lsa, yukning vertikal tebranish davrini aniqlang.

12.24. Prujinaga yuk osilgan. Yuk tebranishining maksimal kinetik energiyasi 1 J ekanligi ma'lum bo'lsa, prujinaning deformatsiya koeffitsientini toping. Tebranish amplitudasi 5 sm.

12.25. Ikkita birday prujinalarga osilgan yukning prujinalarni ketma-ket ulashdan parallel ulashga o'tkazilgandagi vertikal tebranish davri qanday o'zgargan?

12.26. Prujinaga osilgan mis sharcha vertikal tebranyapti. Prujinaga mis sharcha o'miga shunday radiusdagi alyuminiy sharcha osilsa, tebranish davri qanday o'zgaradi?

12.27. Prujinaga toshlar qo'yilgan tarozi pallas osilgan. Bunda vertikal tebranishlar davri 0,5 sek ga teng. Pallaga qo'shimcha toshlar qo'yilganidan keyin vertikal tebranishlar davri 0,6 sek bo'ldi. Shu qo'shimcha yuk tufayli prujina qanchaga cho'zilgan?

12.28. Uzunligi 40 sm va radiusi 1 mm bo'lgan rezina shnurga og'irligi 0,5 kG tosh osilgan. Shu rezinkaning yung moduli $0,3 \text{ kG/mm}^2$ ekanligi ma'lum bo'lsa, toshning vertikal tebranish davrini toping.

Ko'rsatma: Rezinkaning deformatsiya koeffitsientini k rezinkaning yung moduli E bilan $k = \frac{SE}{l}$ nisbatda bog'langanligini e'tiborga oling; bundagi S rezinka ko'ndalang kesimining yuzi va l - uning uzunligi.

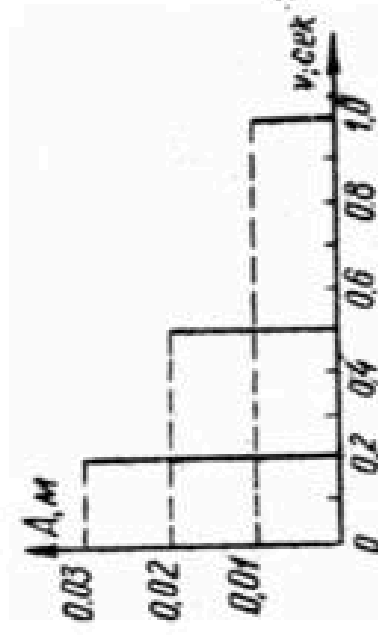
12.29. Og'irligi $P=0,2$ kG bo'lgan areometr suyuqlikda qalqimoqda. Areometrni suyuqlikka bir oz botirib, so'ngra qo'yib yuborisa, $T=3,4$ sek davr bilan tebrana boshlaydi. Tebranishni so'nmaydigan deb hisoblab, shu tajriba ma'lumotlariga asoslanib, areometr qalqib yurgan suyuqlikning zichligi ρ ni toping. Areometr vertikal silindrik naychasining diametri $d=1$ sm.

12.30. 8 sek birday davr va 0,02 m birday amplitudali bir tomonga yo'nalgan ikki garmonik tebranma harakatlarni qo'shish natijasida olingan harakat tenglamasini yozing. Bu tebranishlar fazalarining farqi $\frac{\pi}{4}$ ga teng, birining boshlang'ich fazasi nolga teng.

12.31. $x_1 = 0,02 \sin(5\pi + \frac{\pi}{2})$ m va $x_2 = 0,03 \sin(5\pi + \frac{\pi}{4})$ m tenglamalar bilan berilgan, bir tomonga yo'nalgan tebranishlarni qo'shishdan hosil bo'lgan garmonik tebranish amplitudasi va boshlang'ich fazasini toping.

12.32. Bir tomonga yo'nalgan birday amplituda va davrli ikki garmonik tebranishlarni qo'shish natijasida xuddi shunday davr va amplitudali natijaviy tebranish hosil bo'ladi. Qo'shiluvchi tebranishlar fazalarining farqini toping.

12.33. 1) $x_1 = 4 \sin \pi t$ sm va $x_2 = 3 \sin(\pi t + \frac{\pi}{2})$ sm tenglamalar bilan berilgan bir tomonga yo'nalgan tebranishlarni qo'shishdan hosil bo'lgan garmonik tebranish amplitudasi va boshlang'ich fazasini toping. 2) Natijaviy tebranish tenglamasini yozing. 3) Amplitudalarni qo'shishning vektor diagrammasini chizing.



61-rasm

12.34.61-rasmda murakkab tebranish spektri berilgan. 1) Bu rasm ma'lumotlaridan foydalanib, murakkab tebranishni tashkil etuvchi tebranishlar tenglamasini yozing. 2) Shu tebranishlar grafigini chizing. (t=0 paytda bu tebranishlar fazalarining farqi nolga teng deb qabul qilinadi) 3) Natijaviy murakkab tebranish grafigini chizing.